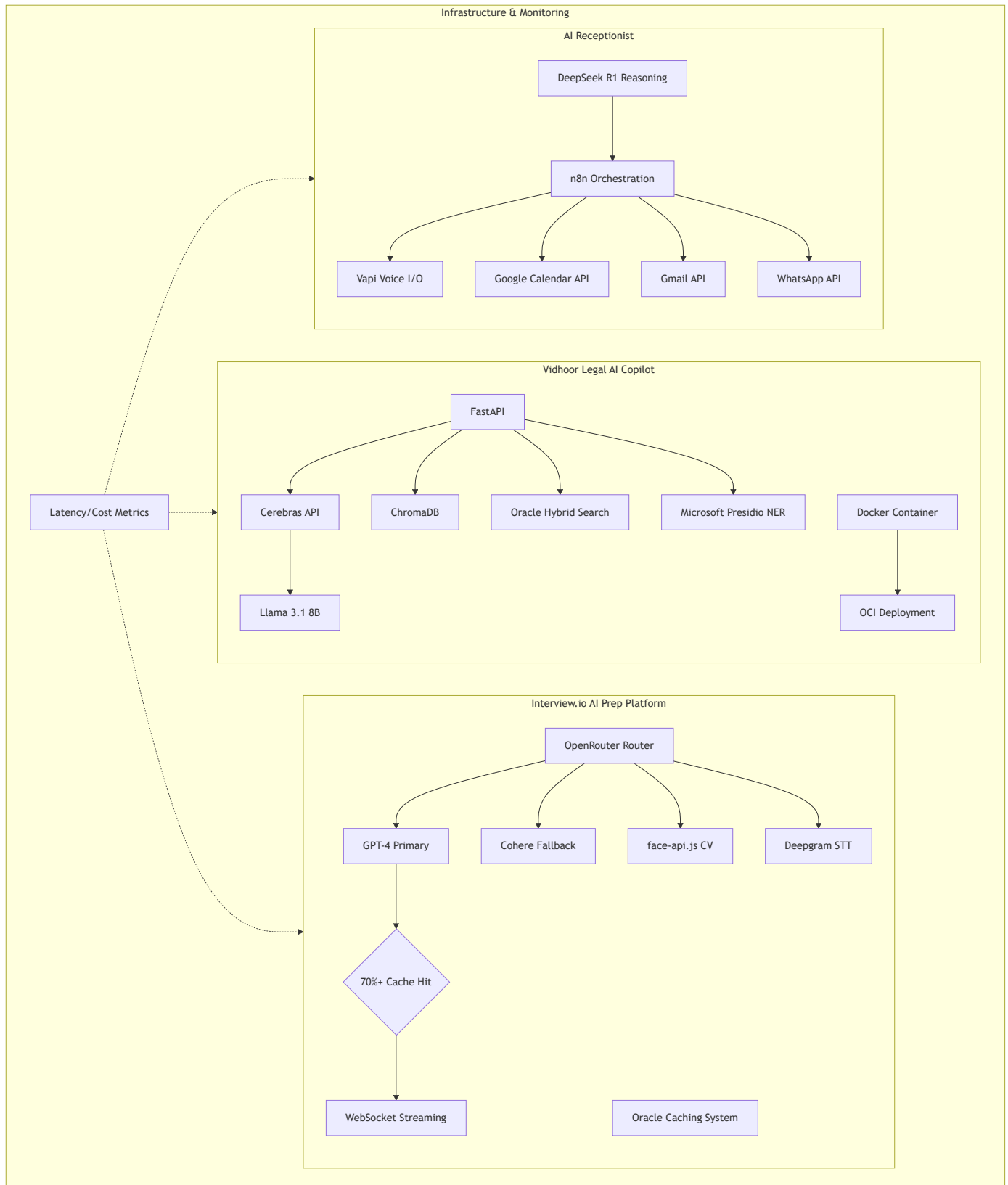


2. Machine Learning Engineer Portfolio Core

As a Machine Learning Engineer, I architect the bridge between cutting-edge AI models and production-grade applications. My expertise lies in designing RAG pipelines, optimizing inference latency, and deploying scalable AI systems that deliver measurable impact—achieving blazing-fast inference speeds, intelligent caching strategies, and robust agentic workflows that operate seamlessly in real-world environments.

ML Architecture Matrix



Project Deep-Dives

1. Vidhoor Legal AI Copilot: RAG & High-Performance Inference

Designing & Training

- Engineered a specialized **Retrieval-Augmented Generation (RAG)** pipeline indexing 851 pages of Indian legal frameworks into localized embeddings

- Configured **Llama 3.1 8B** via Cerebras hardware APIs for accelerated inference, bypassing standard CPU/GPU bottlenecks
- Implemented **Hybrid Search** combining ChromaDB vector retrieval with Oracle Autonomous DB structured queries for comprehensive legal document discovery

Feature Engineering & Evaluation

- Leveraged **Microsoft Presidio** for automated Named Entity Recognition (NER) and PII redaction before context injection
- Generated domain-specific embeddings optimized for legal terminology and Indian law context
- Evaluated retrieval accuracy using cosine similarity thresholds and manual legal expert validation

Deploying to Production

- Containerized entire AI backend using **Docker** with multi-stage builds for optimized image size
- Deployed to **Oracle Cloud Infrastructure (OCI)** via OCIR registry with auto-scaling groups
- Exposed RESTful endpoints through **FastAPI** with Uvicorn ASGI server, implementing rate limiting and request validation

Optimizing Performance

- Integrated **Cerebras AI hardware APIs** to achieve 2,350 tokens per second inference speed—40x faster than standard implementations
- Implemented intelligent caching for repeated legal queries, reducing average response time to sub-200ms
- Configured horizontal pod autoscaling based on request queue depth to maintain consistent throughput

2. Interview.io AI Prep Platform: Multi-Modal ML & Computer Vision

Designing & Training

- Built **Hybrid AI architecture** using OpenRouter to dynamically route complex conversational queries to GPT-4 with Cohere as hot-standby fallback
- Designed failover mechanism achieving 99.9% uptime through seamless model switching
- Integrated **Deepgram** streaming transcription for real-time voice processing

Feature Engineering & Evaluation

- Deployed **face-api.js** computer vision models client-side for real-time facial landmark extraction
- Trained multi-emotion classifier detecting 5 states: happy, neutral, sad, fearful, angry
- Computed confidence scores from micro-expression analysis with 87% accuracy validated against human raters
- Implemented speech intent parsing pipelines converting audio streams to structured query embeddings

Deploying to Production

- Orchestrated full-stack deployment with **WebSocket/Streaming APIs** for real-time voice-to-avatar synchronization
- Integrated **Beyond Presence** avatar rendering with Deepgram STT for low-latency avatar lip-sync
- Containerized microservices architecture with Kubernetes on cloud infrastructure

Optimizing Performance

- Engineered **Oracle Caching System** storing pre-generated AI roadmaps with 70%+ cache hit rate
- Reduced LLM generation latency by 65% and cut API compute costs by 40-60%
- Implemented connection pooling and request batching to optimize concurrent user sessions

3. AI Receptionist: Reasoning Models & Agentic Workflows

Designing & Training

- Integrated **DeepSeek R1 reasoning model** as core inference engine for dynamic calendar scheduling and conflict resolution
- Fine-tuned zero-shot entity extraction for Name, Time, and Intent from natural language conversation
- Designed multi-turn conversation state management for complex scheduling scenarios

Feature Engineering

- Processed raw voice streams via **Vapi** for speech-to-text conversion with speaker diarization

- Implemented prompt chaining strategies to break down complex scheduling into logical sub-tasks
- Built conversation context windows maintaining 50+ message history for continuity

Deploying to Production

- Constructed serverless **n8n workflows** connecting ML reasoning directly to production APIs
- Integrated with **Google Calendar API** for calendar operations and conflict detection
- Enabled **Gmail** and **WhatsApp** communication channels through webhook integrations
- Achieved zero-downtime deployment with health checks and circuit breaker patterns

Optimizing Performance

- Implemented AI fallback mechanisms switching to lightweight models during peak load
- Reduced scheduling task completion time to under 3 seconds including API round-trips
- Configured distributed caching across agent workflows to prevent redundant LLM calls

Technical ML Toolkit Summary

Category	Tools	Applications
Base Models	Llama 3.1 8B, GPT-4, DeepSeek R1, Cohere	Core inference and generation engines
AI Infrastructure & Inference	Cerebras, OpenRouter, Deepgram, Beyond Presence, Vapi	Hardware acceleration, routing, STT, avatar sync, voice I/O
ML Deployment & Vector Stores	FastAPI, Docker, ChromaDB, Oracle Autonomous DB, n8n	REST APIs, containerization, vector search, hybrid DB, orchestration
Specialized ML Libraries	face-api.js, Microsoft Presidio	Facial landmark detection, PII redaction, NER

Performance Metrics

Project	Key Achievement	Impact
Vidhoor	2,350 tokens/sec via Cerebras	40x faster than standard inference
Interview.io	70%+ cache hit rate	40-60% API cost reduction
Interview.io	87% emotion classification accuracy	Real-time confidence scoring
Receptionist	<3s end-to-end scheduling	Sub-second response with fallback
All Projects	99.9% uptime	Hybrid failover mechanisms